



CS7.404: Digital Image Processing

Monsoon 2025: Image Restoration



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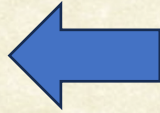
Biometrics and Secure ID Lab, CVIT,
IIIT Hyderabad



Image Restoration



Original Image



Blurred Image

- What caused the blurring?
 - Camera: Translation, shake, out-of-focus, Lens aberrations.
 - Environment: Light scattering, reflection
 - Device: Sensor noise; Quantization

Can we undo these effects to "restore" the image to its original state?



Degradations



- Original



- Optical blur



- Motion blur



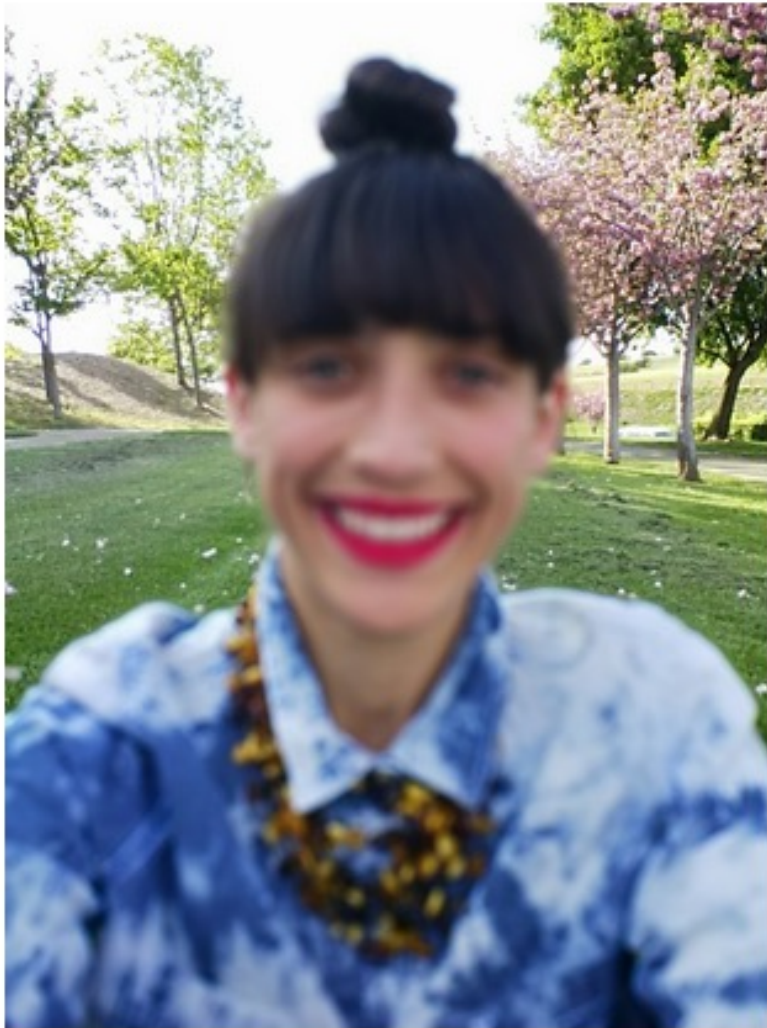
- Spatial quantization (discrete pixels)



- Additive intensity noise



Examples (Optical Blur)



Lens Blur selfie, background focus



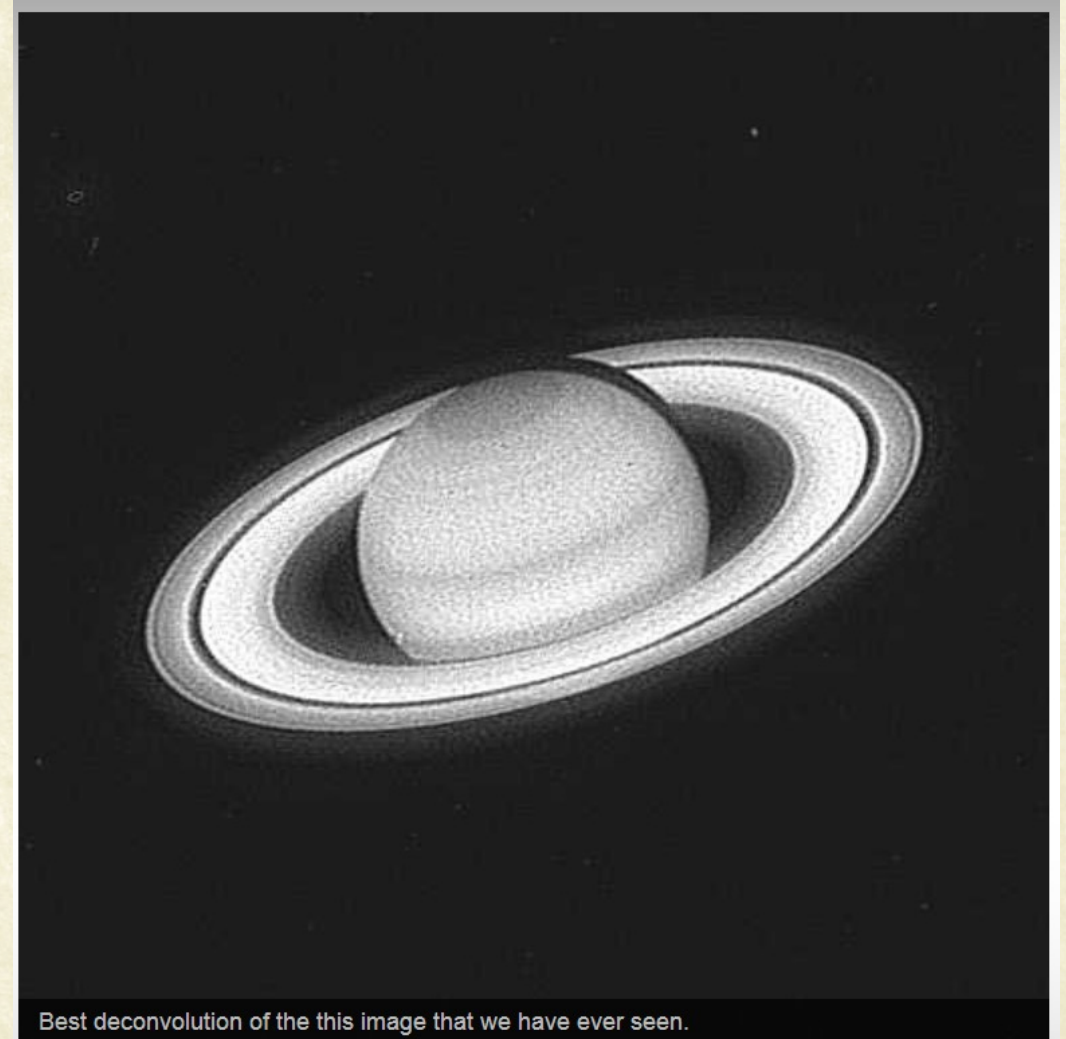
Photo by Rachel Been

Lens Blur selfie, foreground focus

Interesting read:
Light Field Cameras



Examples (Optical Blur)



Courtesy: NASA



Examples (Restoration from camera shake)





Examples (Atmospheric conditions)



Single Image Haze Removal [He et al. CVPR 2009]



Image Restoration

- Started from 1950s
- Application Domains
 - Scientific explorations
 - Legal investigations
 - Film making and archival
 - Image and video de-coding
 - Consumer photography ...
- Related Problem:
 - Image reconstruction in radio astronomy, radar imaging and tomography.

Example of image restoration
Asteroid Vesta

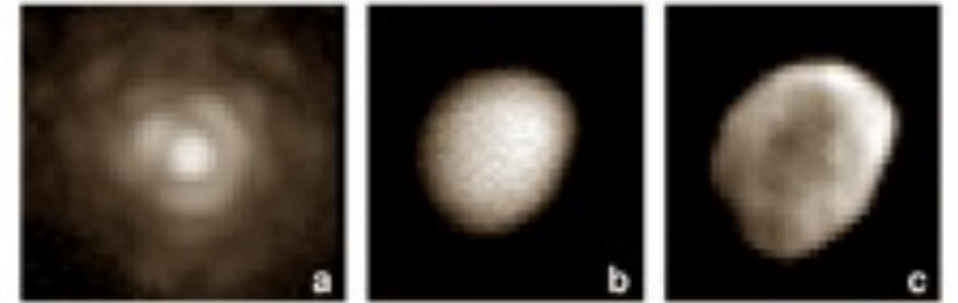


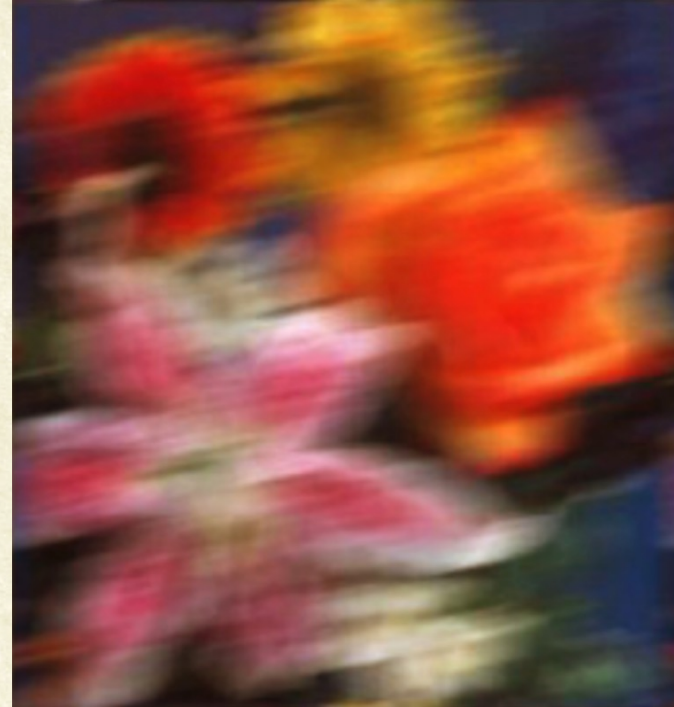


Image Enhancement

- “Improve” the appearance of an image; a subjective process.



Original Image



Blurred Image

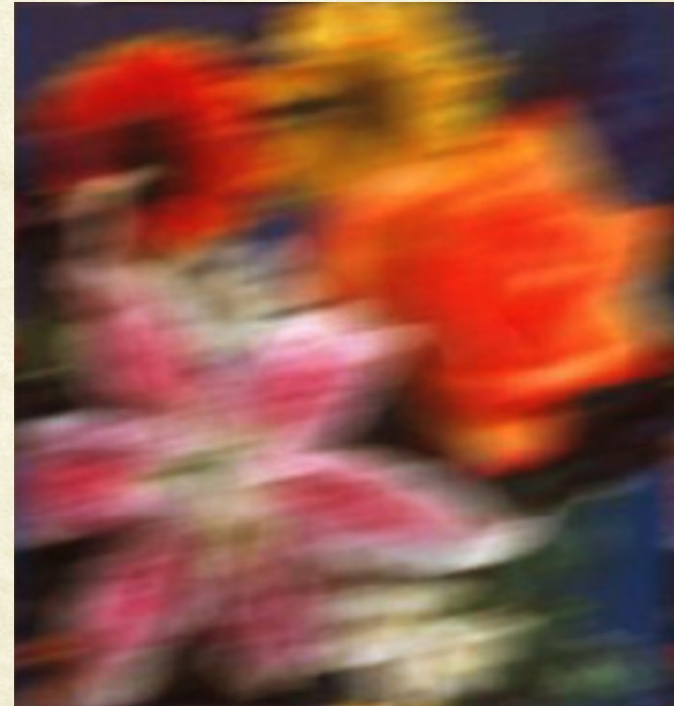
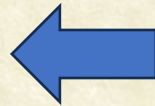


Image Restoration

- Remove distortions from an image to go back to the “original” image; an objective process.



Original Image



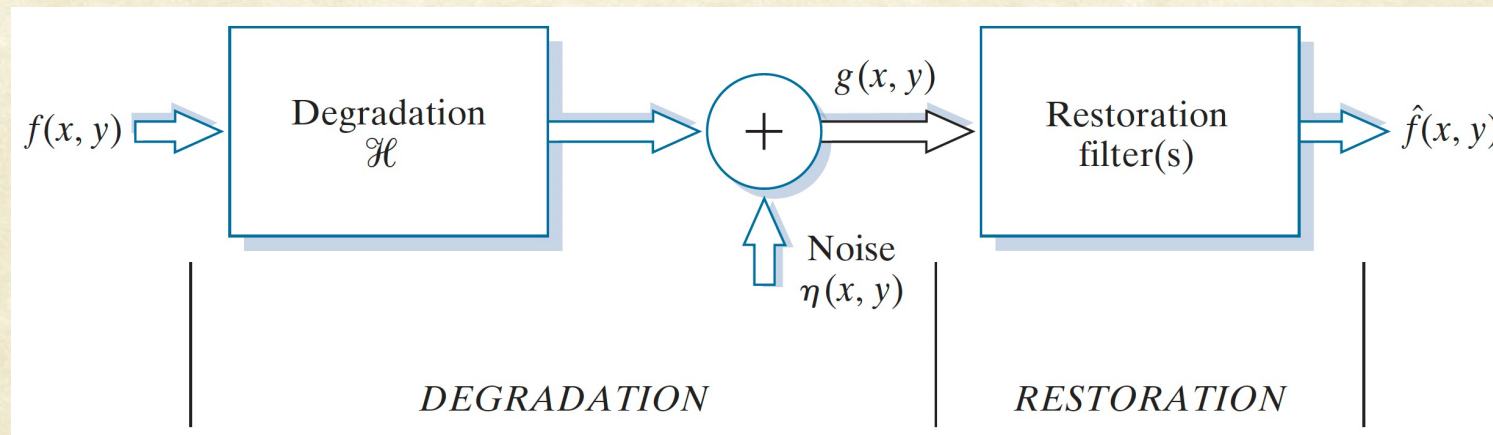
Degraded Image



A Model for Image Degradation and Restoration

- Image Restoration

- Use a priori knowledge of the degradation
- Modeling the degradation and apply the inverse process
- Formulate and evaluate objective criteria of goodness



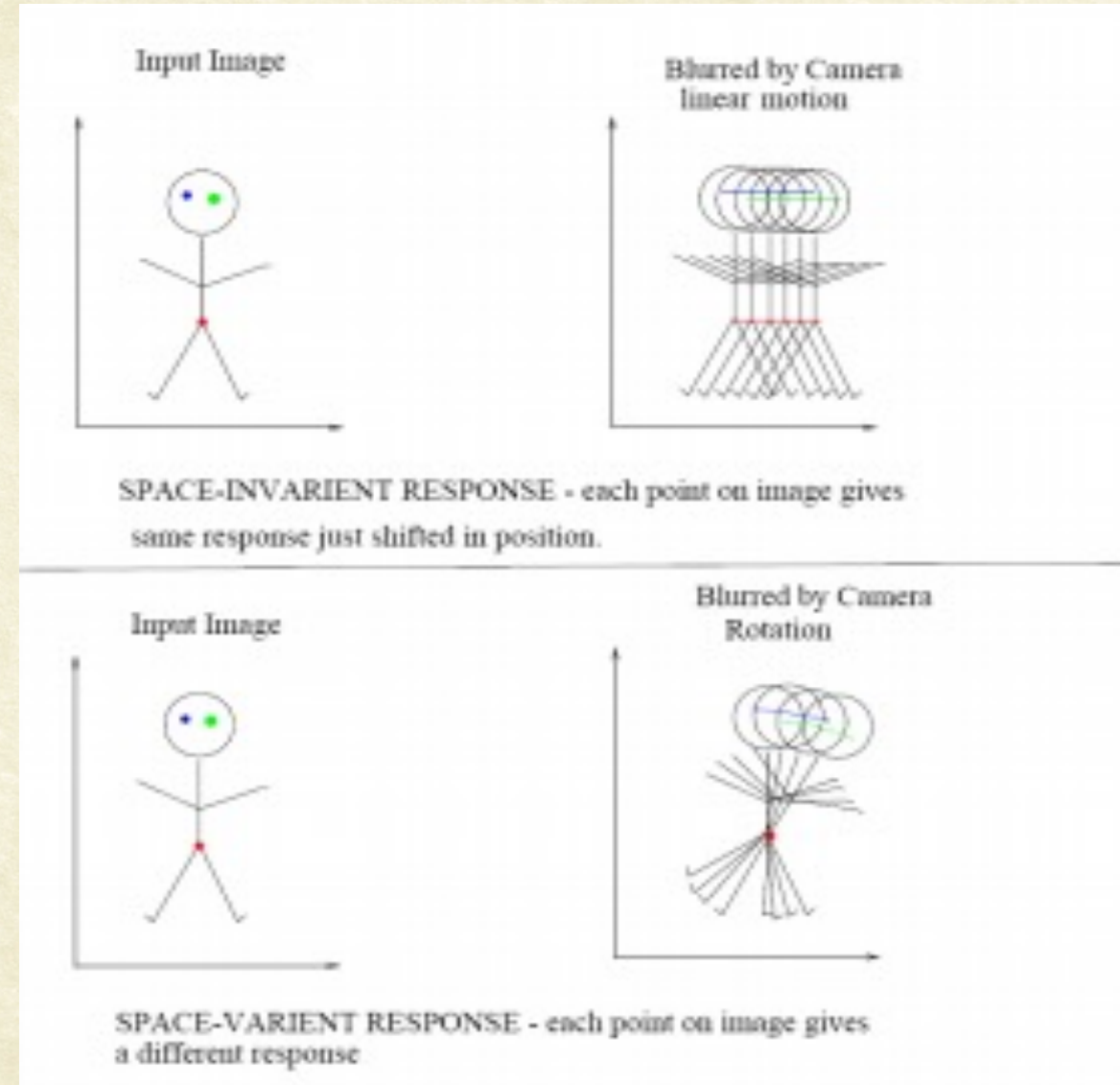
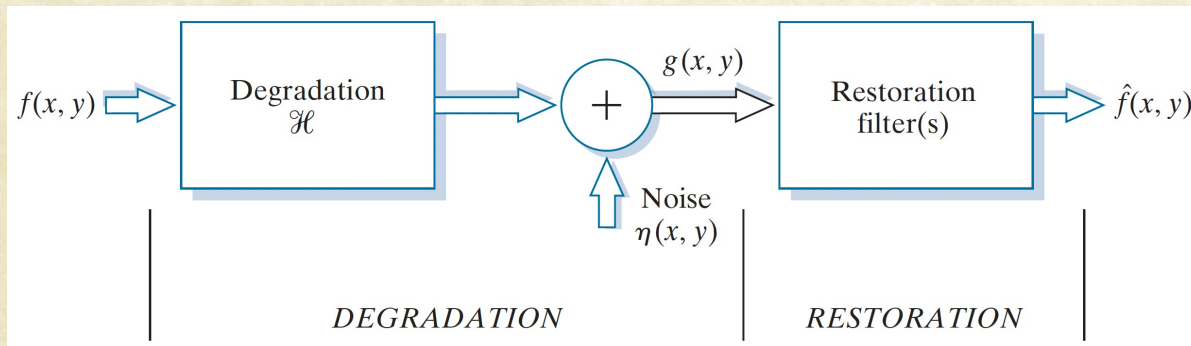
$$g(x, y) = H[f(x, y)] + \eta(x, y)$$

- Design the restoration filters such that $\hat{f}(x, y)$ is as close to $f(x, y)$ as possible.



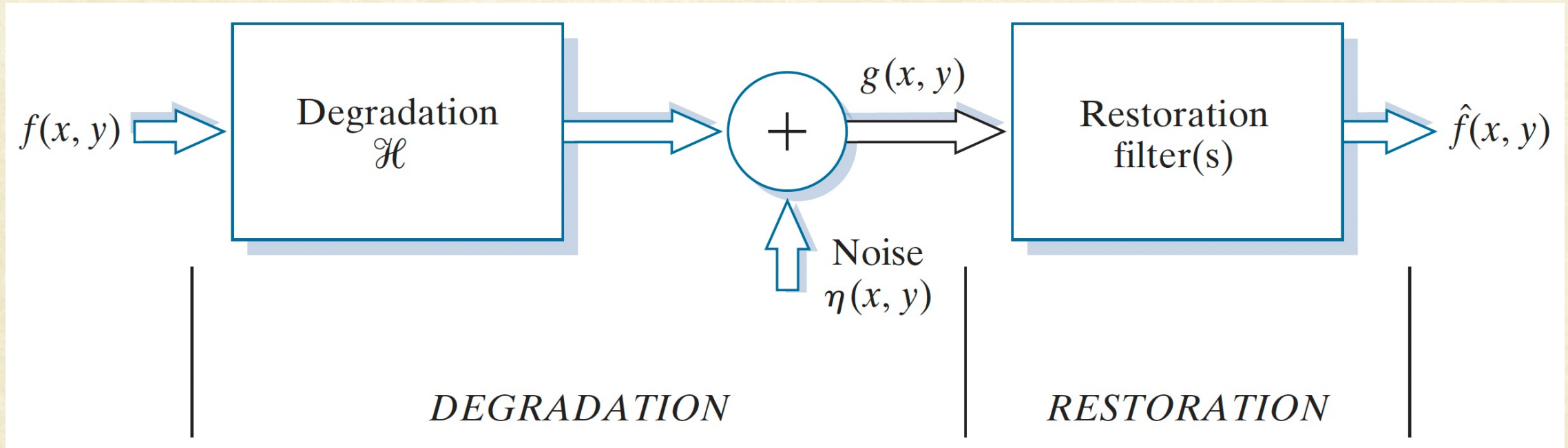
Assumptions for the Distortion Model

- Noise
 - Independent of spatial location
 - except for periodic noise
 - Uncorrelated with the image
- Degradation function, H
 - Linear
 - Position-invariant





Mathematical model of Image Degradation/Restoration



$$g(x, y) = H[f(x, y)] + \eta(x, y)$$

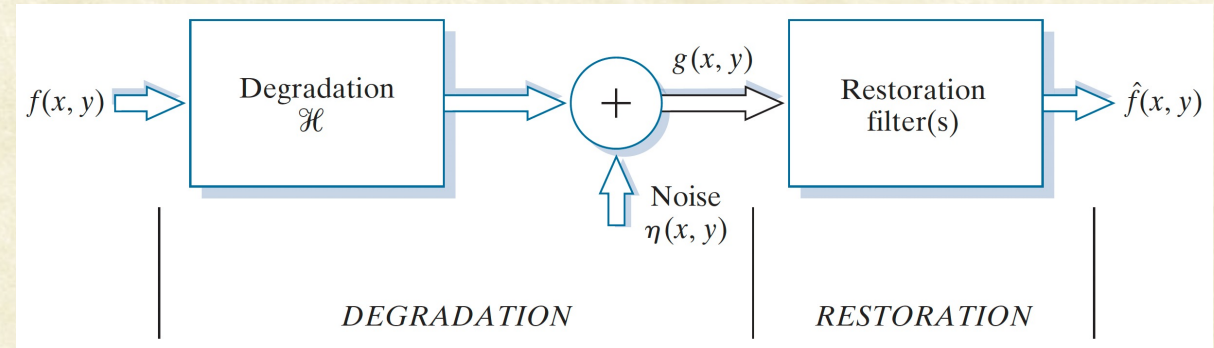
If H is a linear, position-invariant process,

$$g(x, y) = h(x, y) \star f(x, y) + \eta(x, y), \text{ or}$$
$$G(u, v) = H(u, v) F(u, v) + N(u, v)$$



Divide and Conquer: Step #1

- Image degraded only by noise



- Assuming 'H' is identity, the model reduces to:

$$g(x, y) = f(x, y) + \eta(x, y)$$

or

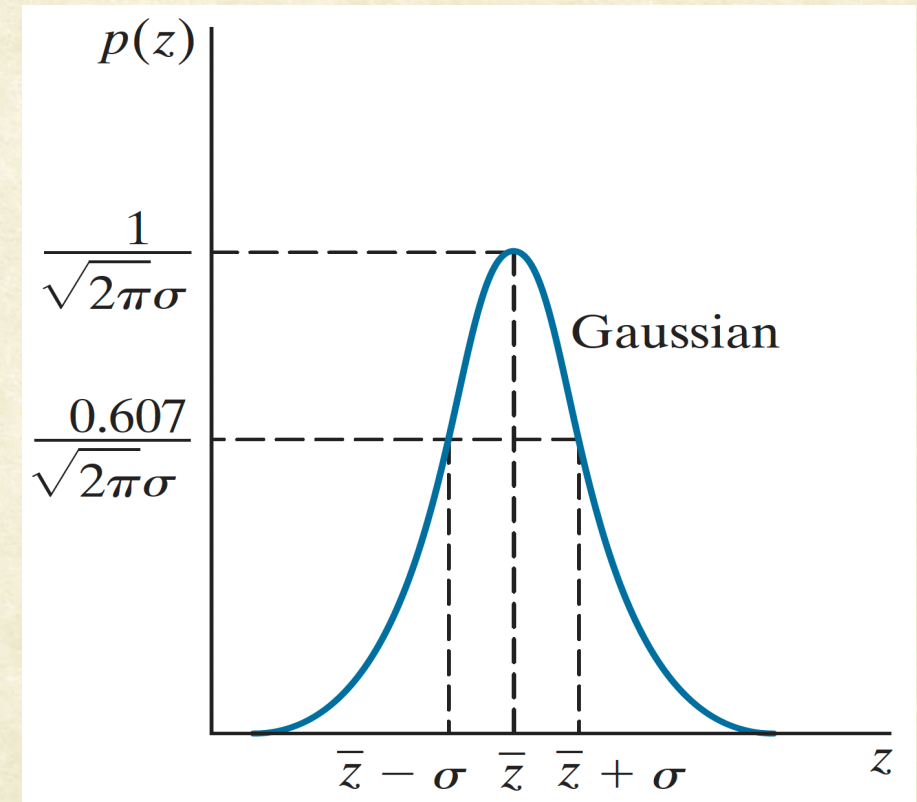
$$G(u, v) = F(u, v) + N(u, v)$$



Noise Models: Gaussian

- Gaussian (normal) Noise.
- Widely used due to
 - mathematical convenience
 - Robustness to model error

$$p(z) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(z-\bar{z})^2}{2\sigma^2}}$$

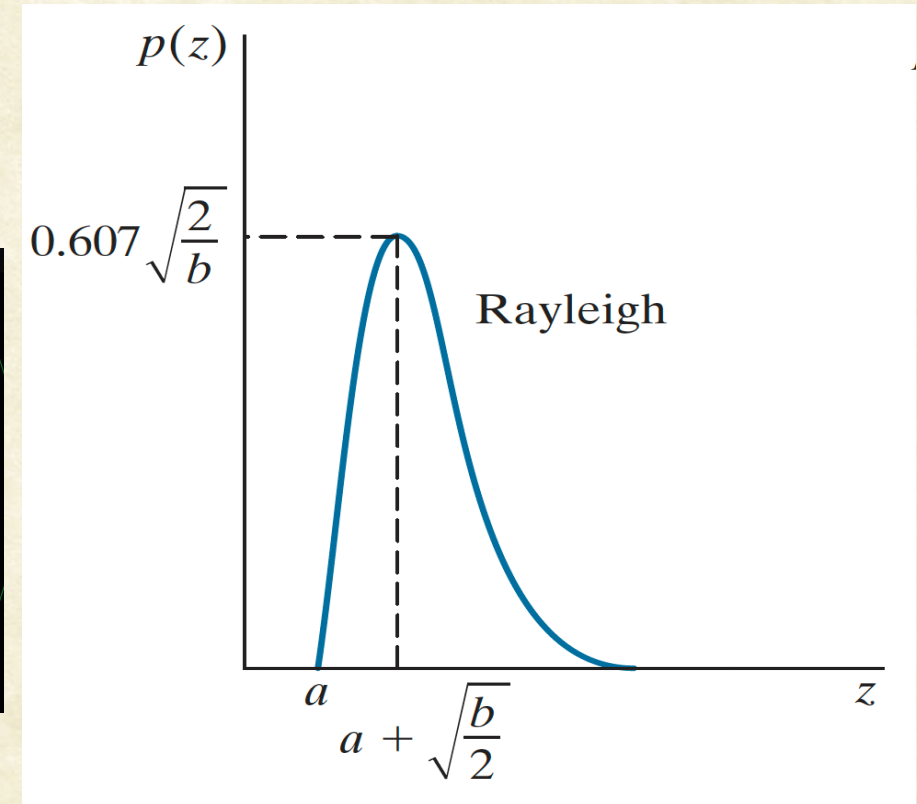
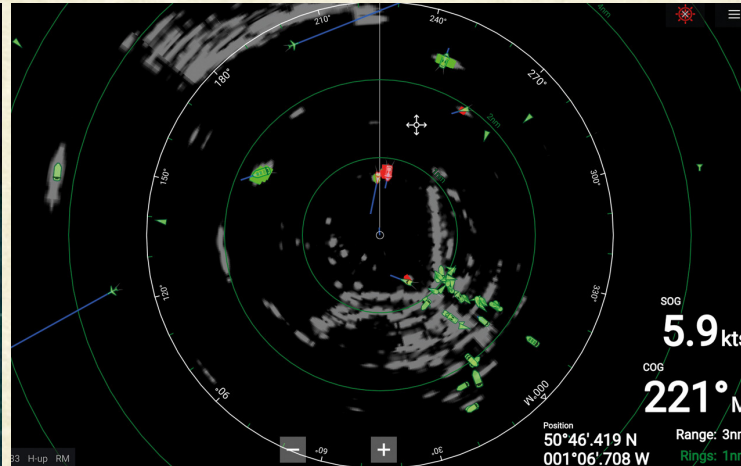
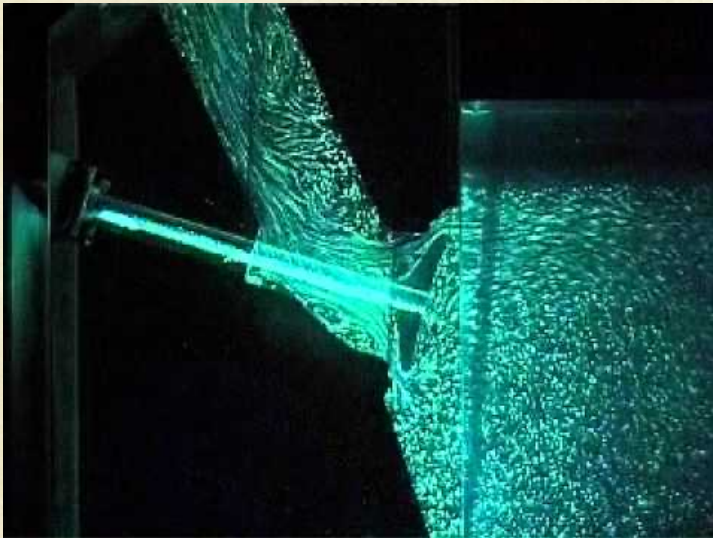




Noise Models: Rayleigh

$$p(z) = \begin{cases} \frac{2}{b}(z - a)e^{-\frac{(z-a)^2}{b}} & z \geq a \\ 0 & z < a \end{cases}$$

- Rayleigh Noise
 - Radar images
 - Velocity images



$$\bar{z} = a + \sqrt{\pi b/4}$$

$$\sigma^2 = \frac{b(4 - \pi)}{4}$$



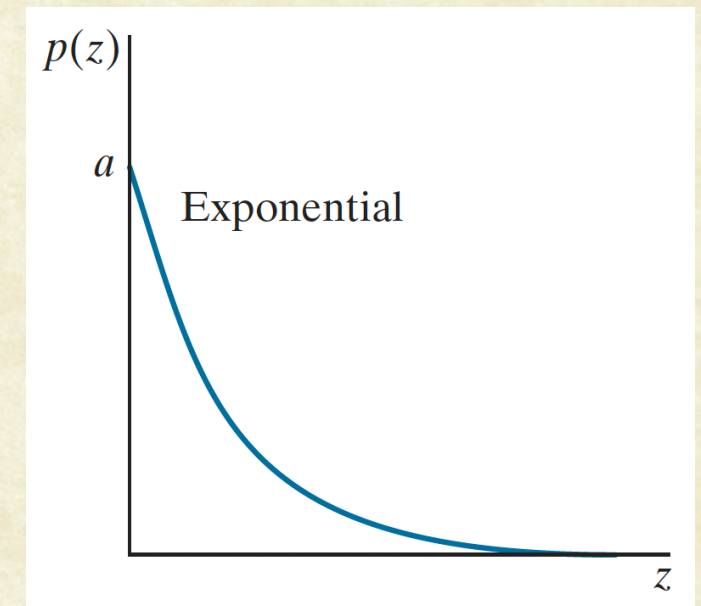
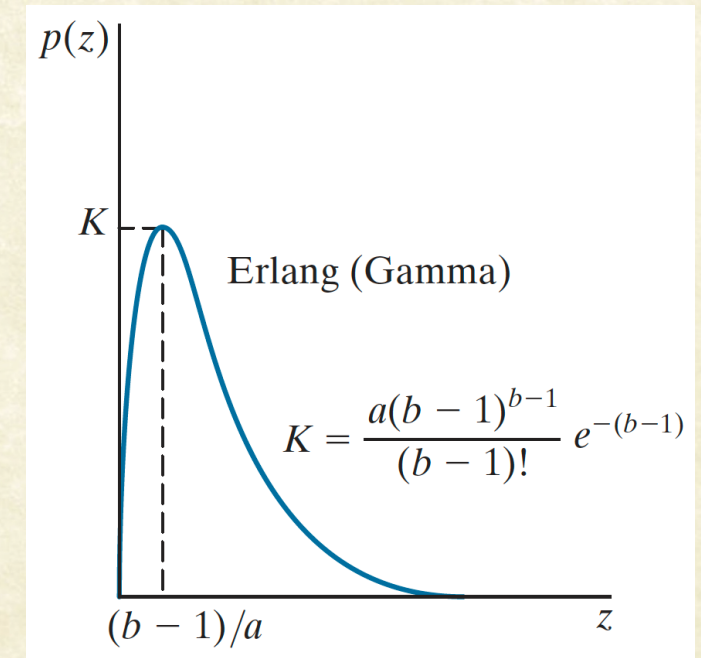
Other Noise Models

- Erlang (Gamma) Noise [Laser images]

$$p(z) = \begin{cases} \frac{a^b z^{b-1}}{(b-1)!} e^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$

- Exponential Noise

$$p(z) = \begin{cases} ae^{-az} & z \geq 0 \\ 0 & z < 0 \end{cases}$$





Noise Models

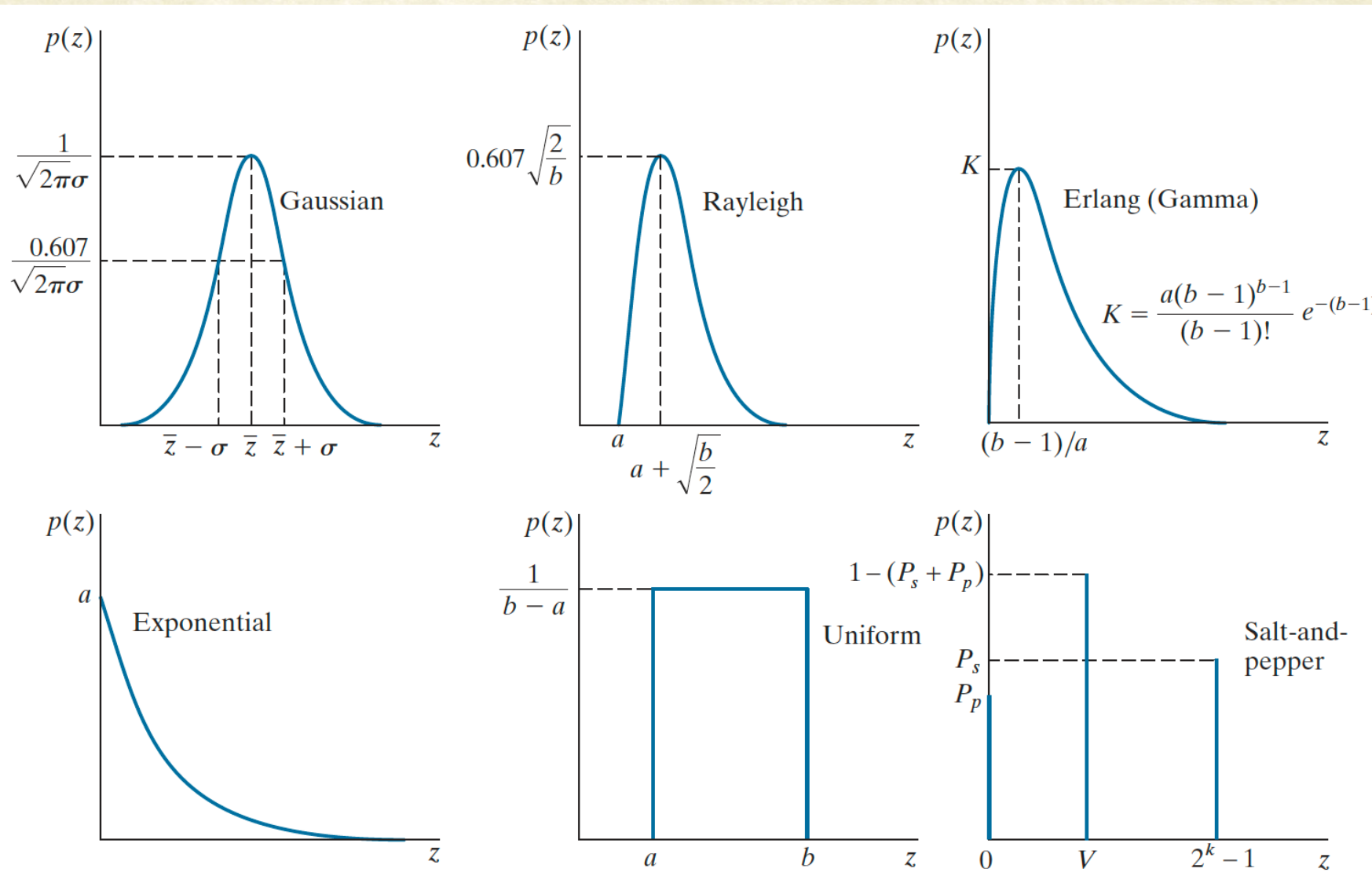
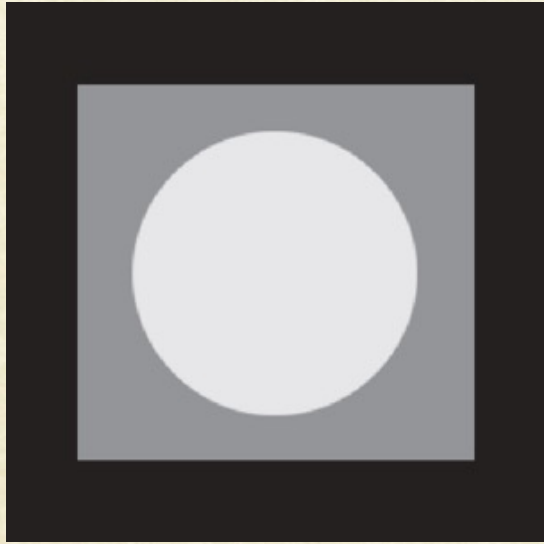
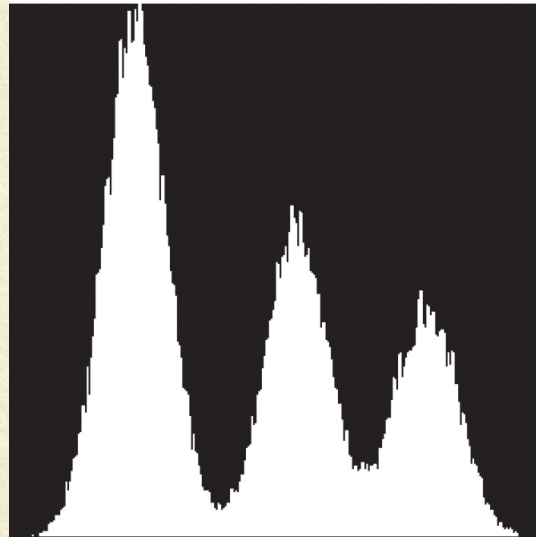
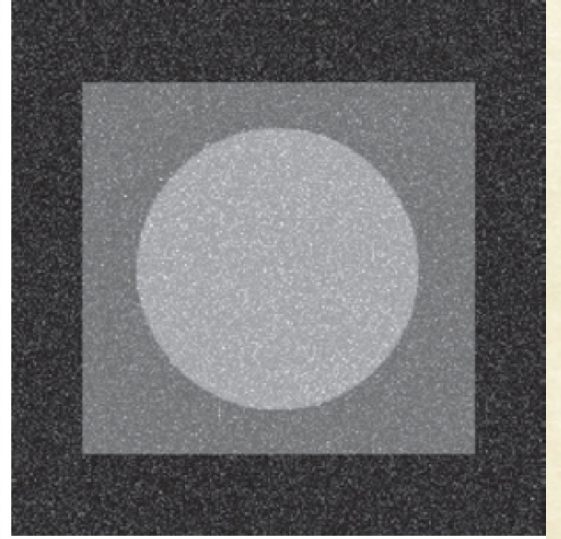
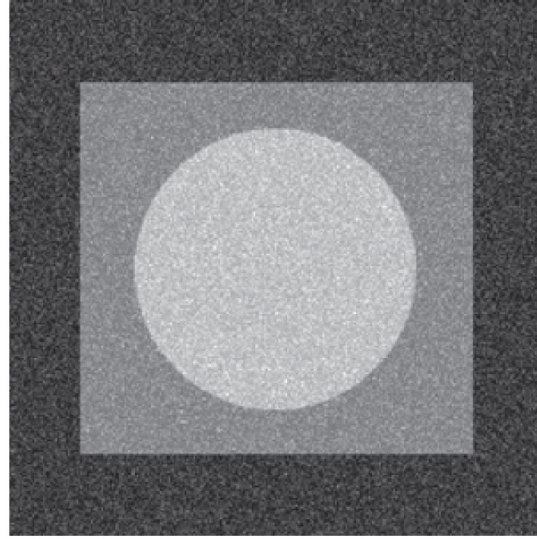
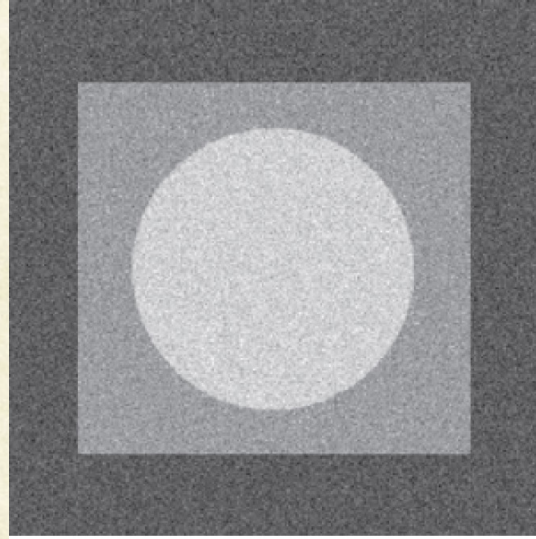




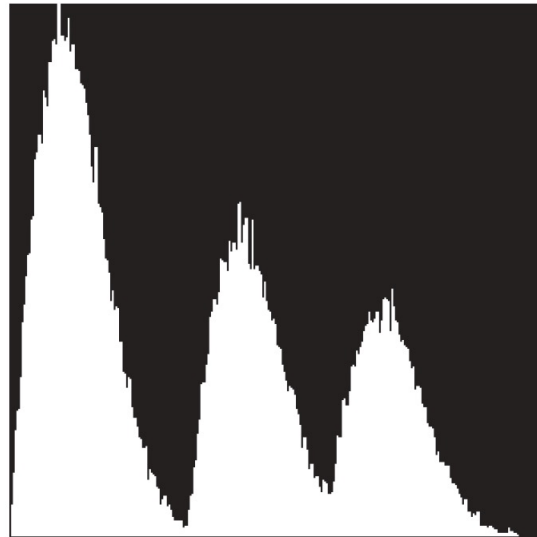
Illustration of Noise Models



Original Image



Gaussian



Rayleigh

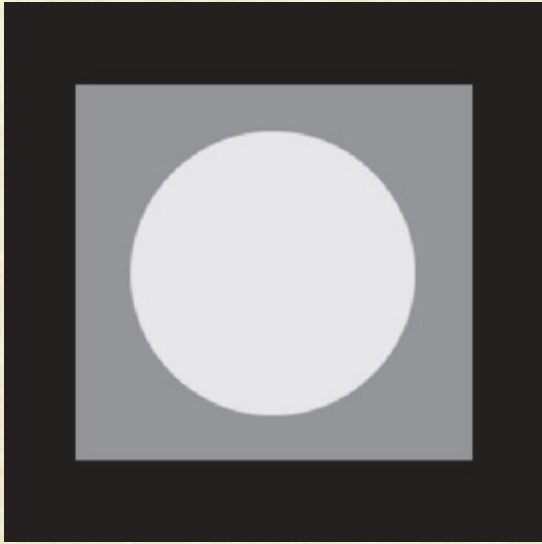


Erlang

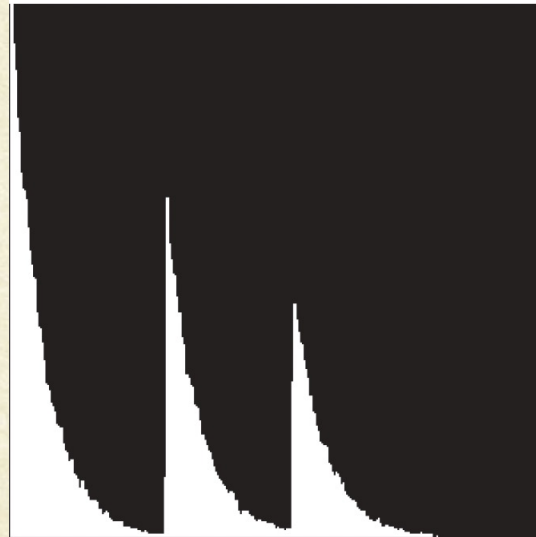
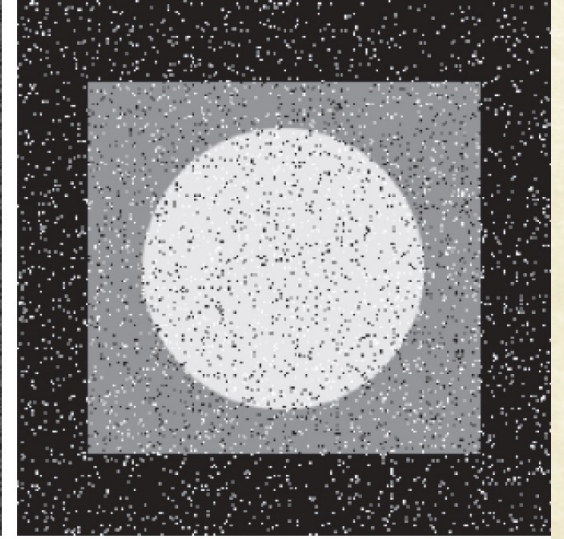
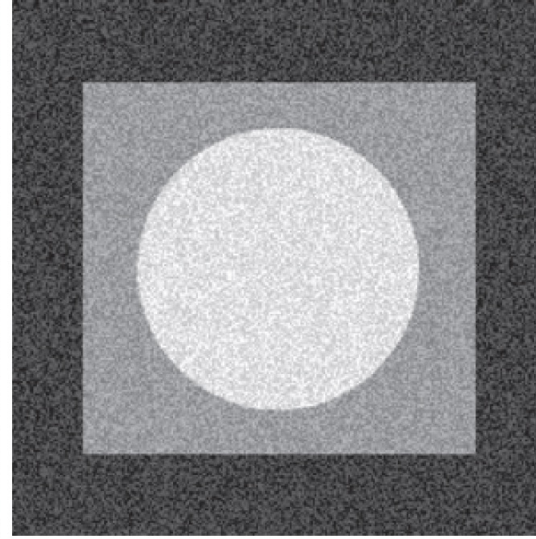
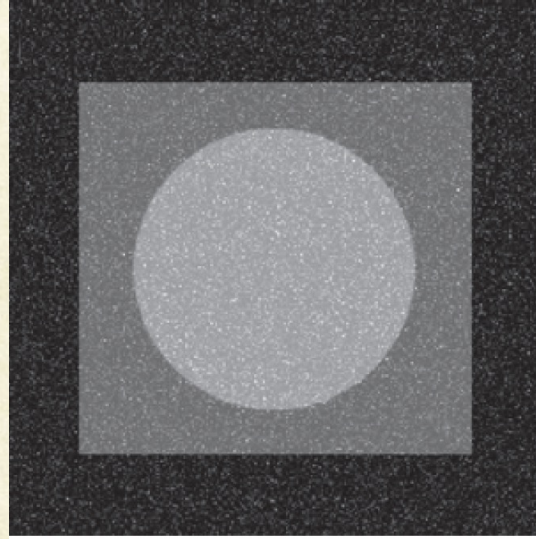
- Visually similar.
- Not easy to determine noise model from appearance



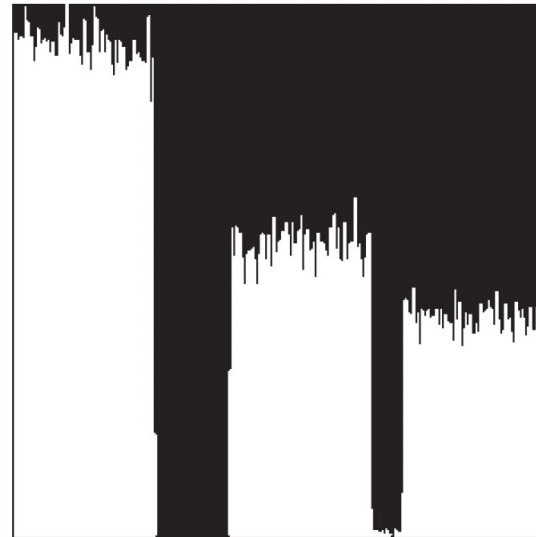
Illustration of Noise Models



Original Image



Exponential



Uniform



Salt & Pepper

- Visually similar.
- Not easy to determine noise model from appearance



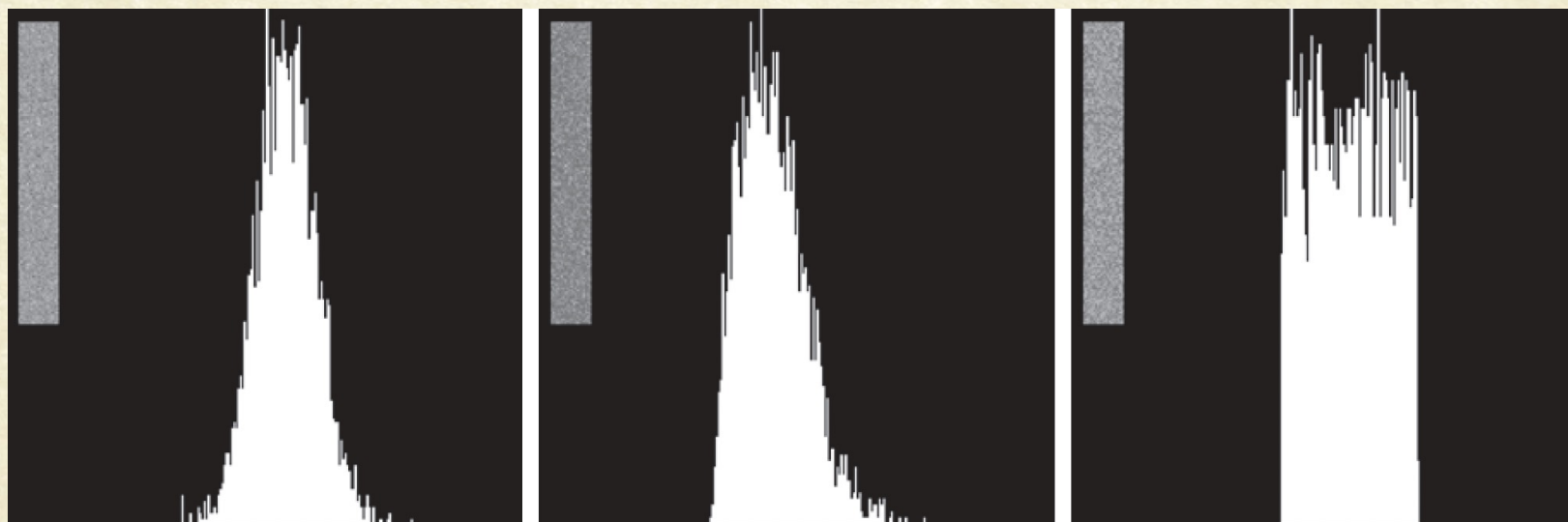
Understanding System Noise

- Case 1: Imaging system available
 - Noise Calibration: Capture a set of 'flat environments' (e.g. solid gray board, object at fixed location)
 - Select the model with better statistical test scores (Akaike Information Criteria (AIC) or Likelihood Ratio Test (LRT))
 - Compute model parameters (mean, variance, etc.) from the statistics of pixel values.



Understanding System Noise

- Case 2: Only images available
 - Estimate from patches of constant intensity
 - For impulse noise, use a mid-gray patch/area



Gaussian

Rayleigh

Uniform



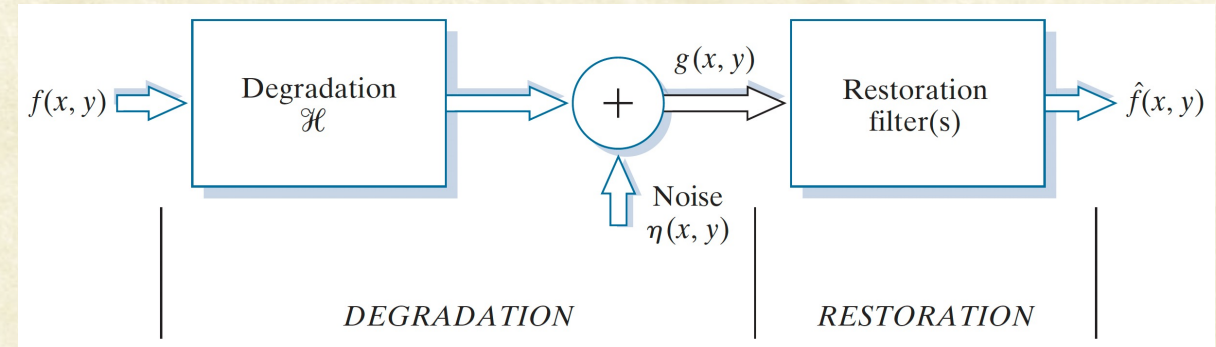
Restoration (in presence of noise only)

- Assuming 'H' is identity, the model reduces to:

$$g(x, y) = f(x, y) + \eta(x, y)$$

or

$$G(u, v) = F(u, v) + N(u, v)$$





Restoration (in presence of noise only)

- Mean filters

- Arithmetic mean filter

$$\hat{f}(x, y) = \frac{1}{mn} \sum_{(r, c) \in S_{xy}} g(r, c)$$

- Geometric mean filter

$$\hat{f}(x, y) = \left[\prod_{(r, c) \in S_{xy}} g(r, c) \right]^{\frac{1}{mn}}$$



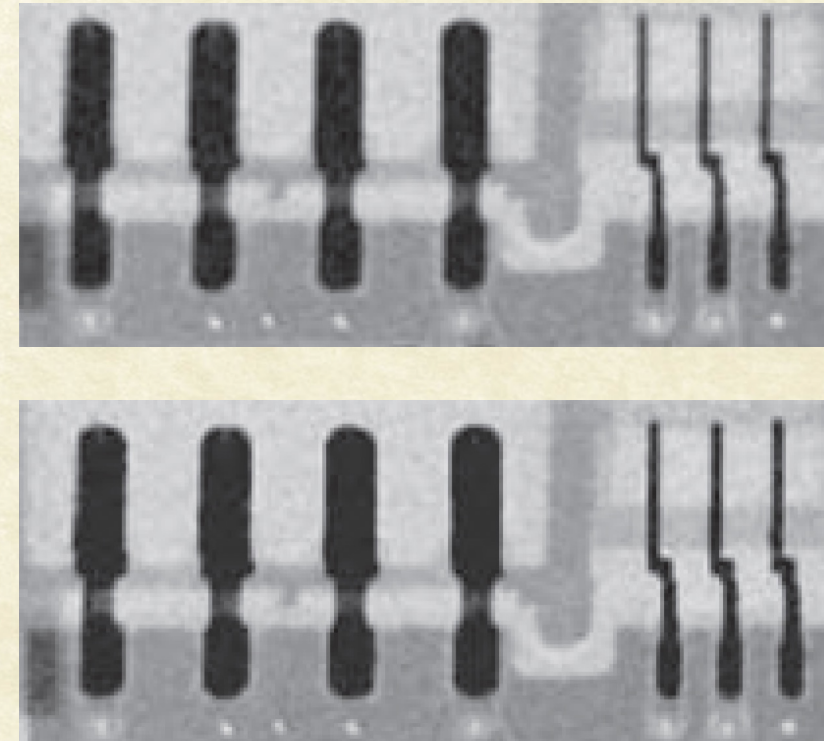
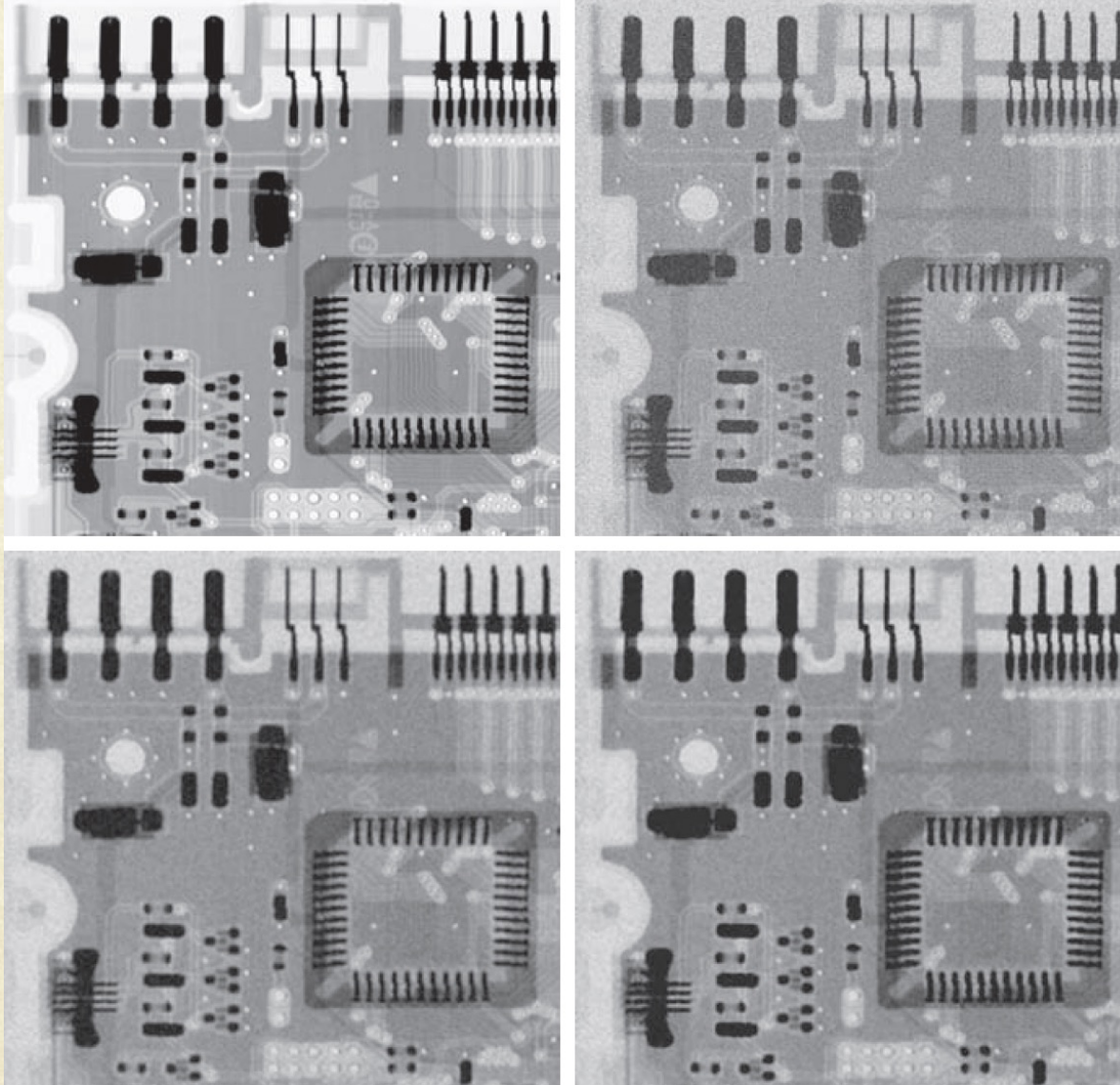


Restoration (in presence of noise only)

a	b
c	d

FIGURE 5.7

(a) X-ray image of circuit board. (b) Image corrupted by additive Gaussian noise. (c) Result of filtering with an arithmetic mean filter of size 3×3 . (d) Result of filtering with a geometric mean filter of the same size. (Original image courtesy of Mr. Joseph E. Pascente, Lixi, Inc.)





Restoration (in presence of noise only)

- Mean filters
 - Harmonic mean filter (Gaussian, Salt)

$$\hat{f}(x, y) = \frac{mn}{\sum_{(r,c) \in S_{xy}} \frac{1}{g(r, c)}}$$

- Contraharmonic mean filter (Salt & Pepper)

$$\hat{f}(x, y) = \frac{\sum_{(r,c) \in S_{xy}} g(r, c)^{Q+1}}{\sum_{(r,c) \in S_{xy}} g(r, c)^Q}$$

Q : order of the filter; Good for salt and pepper noise

Eliminates pepper noise for $Q > 0$ and salt noise for $Q < 0$

NB: cf. arithmetic filter if $Q = 0$, harmonic mean if $Q = -1$

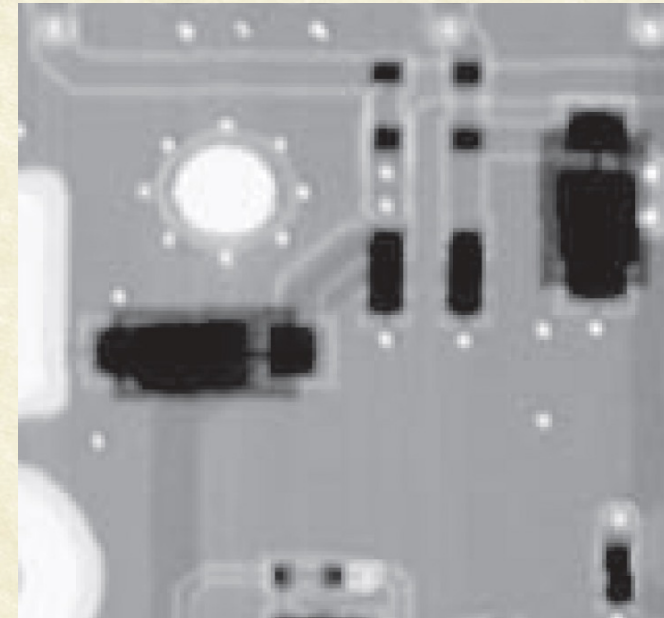
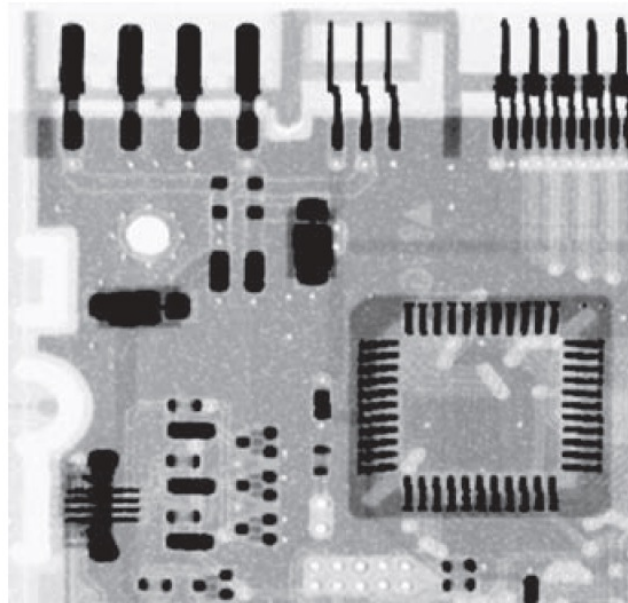
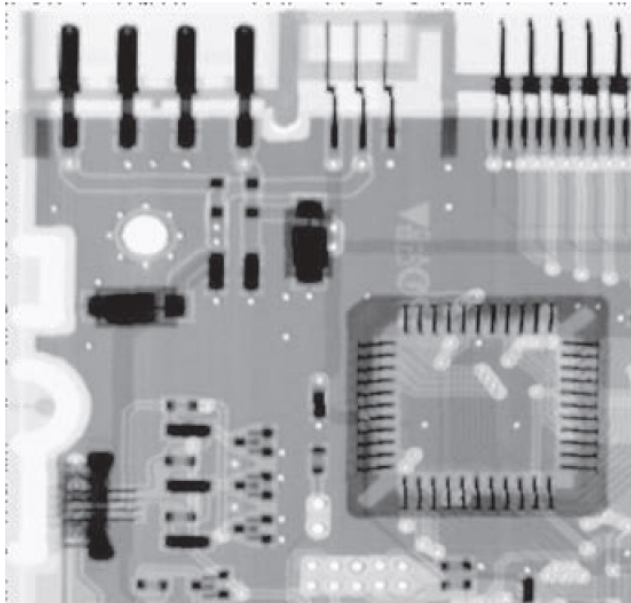
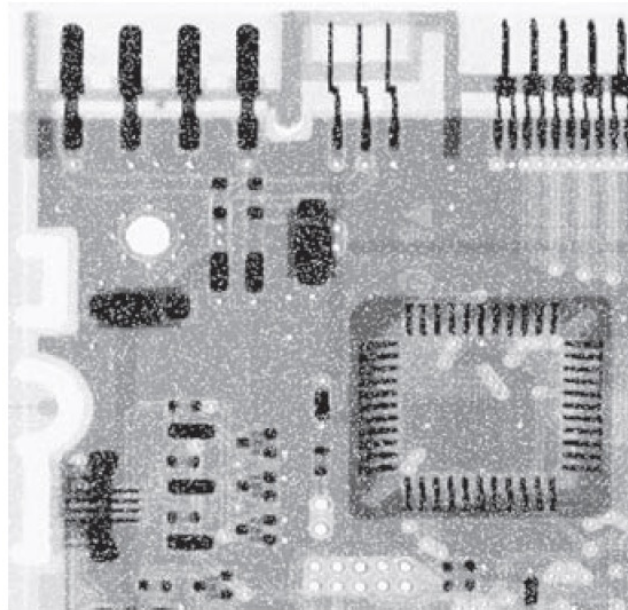
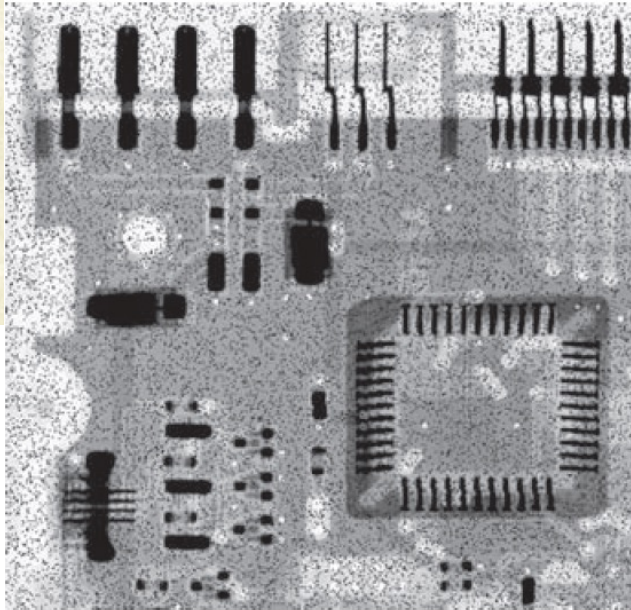


Restoration (in presence of noise only)

a	b
c	d

FIGURE 5.8

(a) Image corrupted by pepper noise with a probability of 0.1. (b) Image corrupted by salt noise with the same probability. (c) Result of filtering (a) with a 3×3 contra-harmonic filter $Q = 1.5$. (d) Result of filtering (b) with $Q = -1.5$.





Restoration (in presence of noise only)

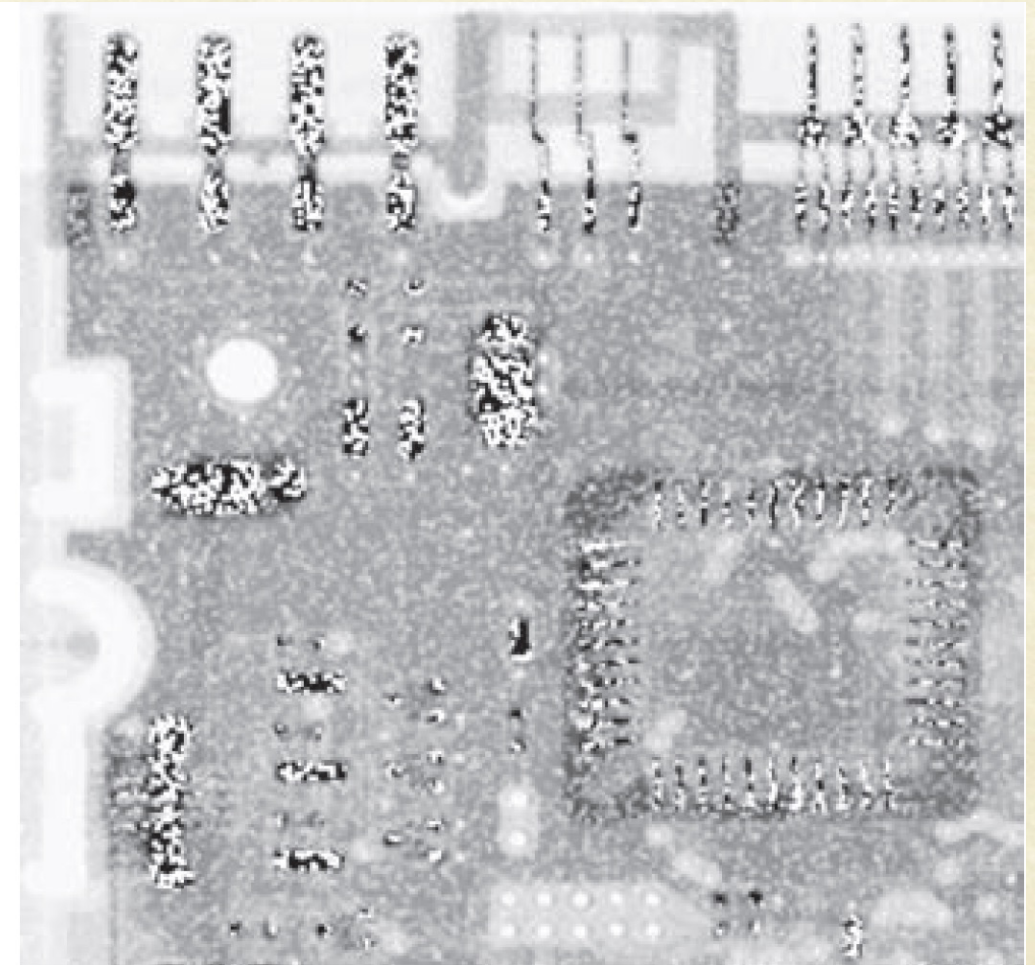
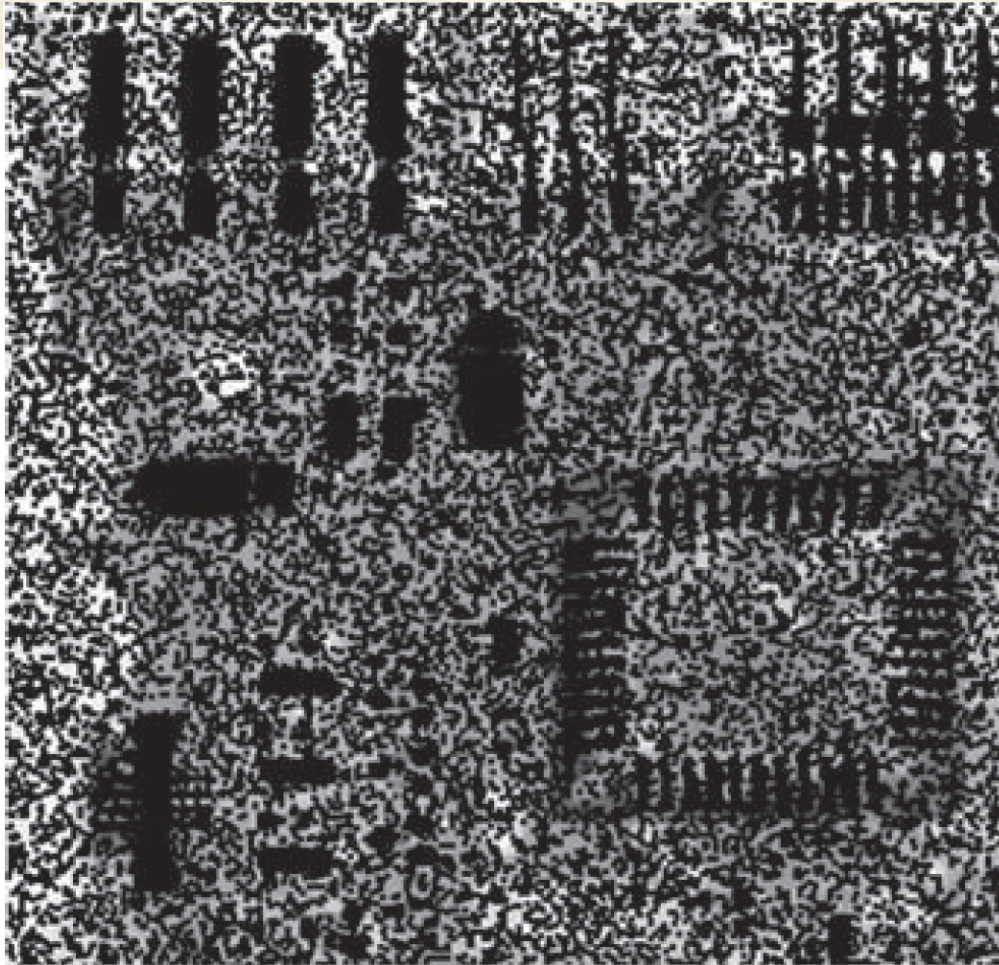
a b

FIGURE 5.9

Results of selecting the wrong sign in contraharmonic filtering.

(a) Result of filtering Fig. 5.8(a) with a contraharmonic filter of size 3×3 and $Q = -1.5$.

(b) Result of filtering Fig. 5.8(b) using $Q = 1.5$.





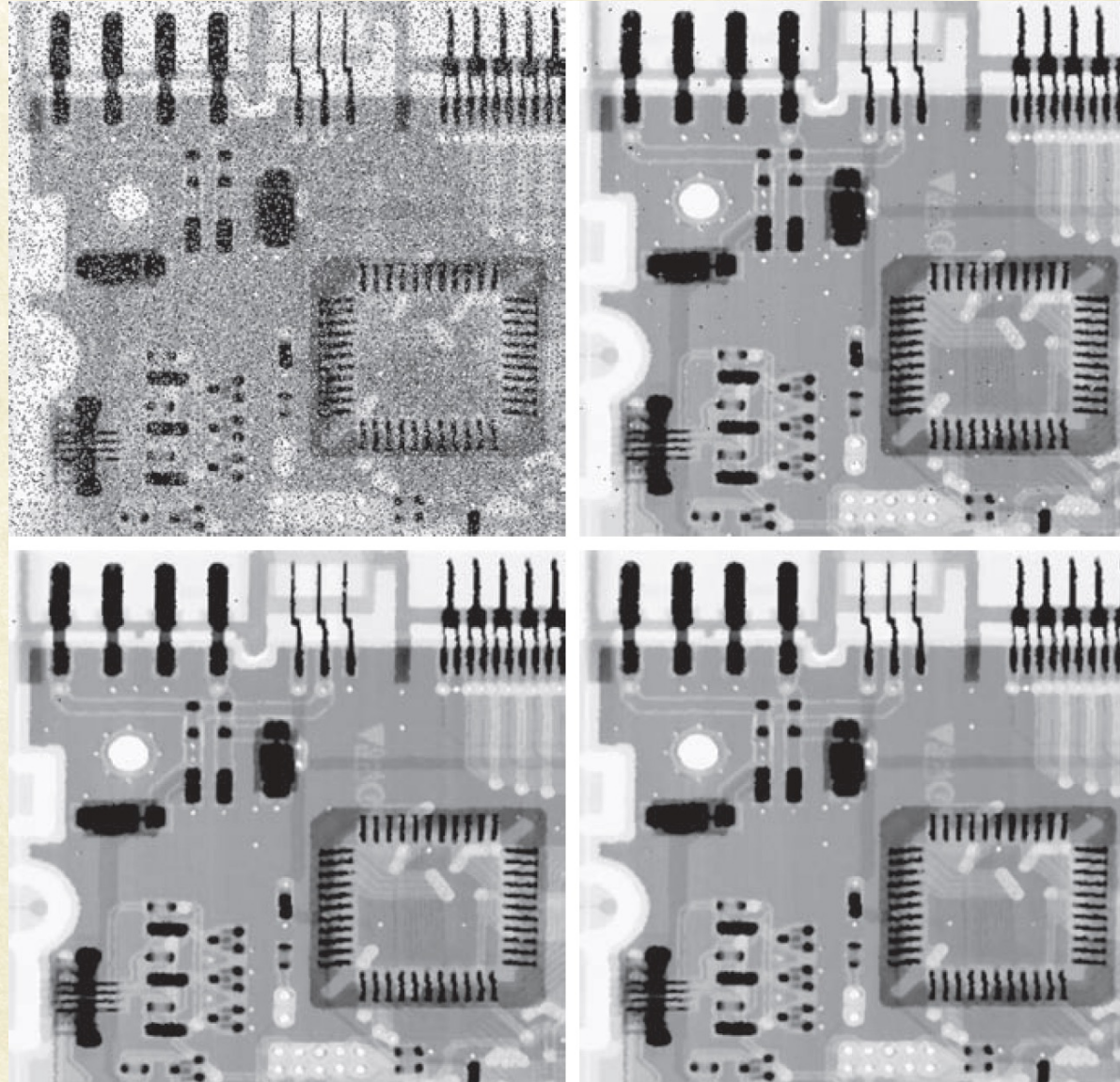
Restoration (in presence of noise only)

- Median filter

a	b
c	d

FIGURE 5.10

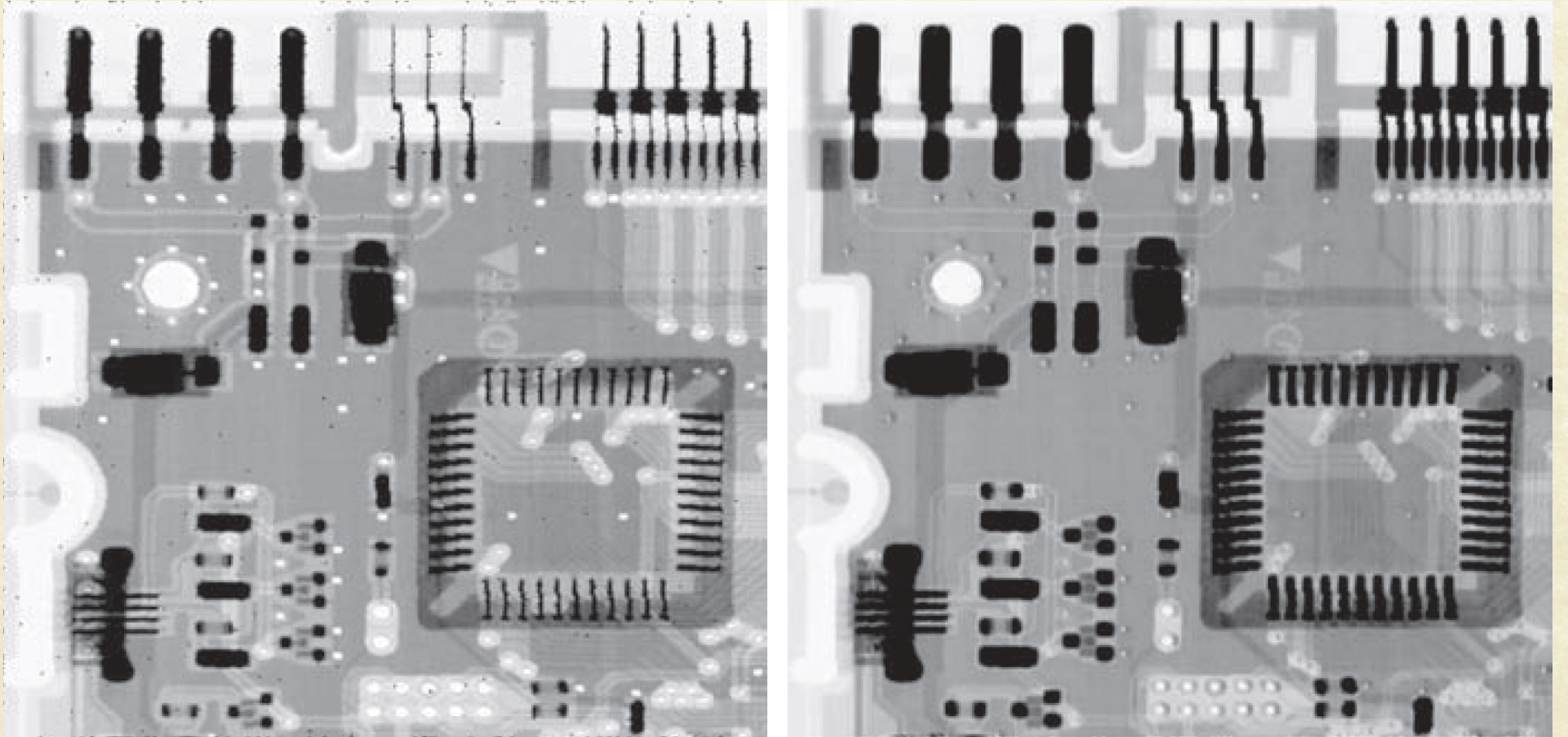
(a) Image corrupted by salt-and-pepper noise with probabilities $P_s = P_p = 0.1$. (b) Result of one pass with a median filter of size 3×3 . (c) Result of processing (b) with this filter. (d) Result of processing (c) with the same filter.





Restoration (in presence of noise only)

- Max, Min filters



a b

FIGURE 5.11

(a) Result of filtering Fig. 5.8(a) with a max filter of size 3×3 .

(b) Result of filtering Fig. 5.8(b) with a min filter of the same size.



Restoration (in presence of noise only)

- Midpoint filter

$$\hat{f}(x, y) = \frac{1}{2} \left[\max\{g(s, t)\}_{(s, t) \in S_{xy}} + \min\{g(s, t)\}_{(s, t) \in S_{xy}} \right]$$

Best for
Uniform
or
Gaussian
noise

- Alpha trimmed filter

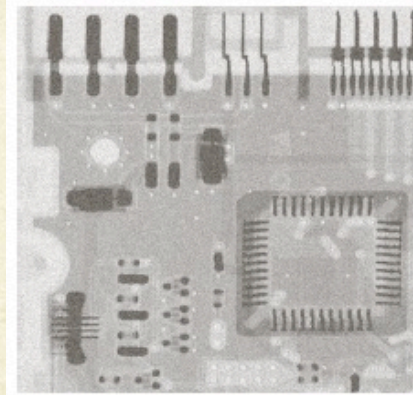
$$\hat{f}(x, y) = \frac{1}{mn - d} \sum_{(s, t) \in S_{xy}} g_r(s, t)$$

Where g_r represents the image g in which the $d/2$ lowest and $d/2$ highest intensity values in the neighbourhood S_{xy} were deleted

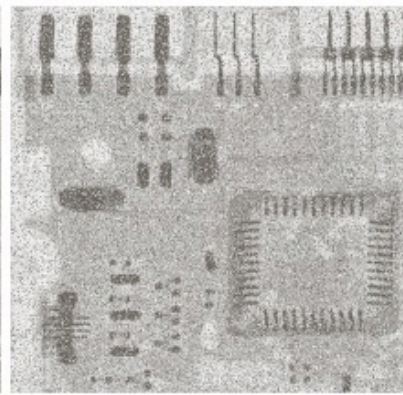


Restoration (in presence of noise only)

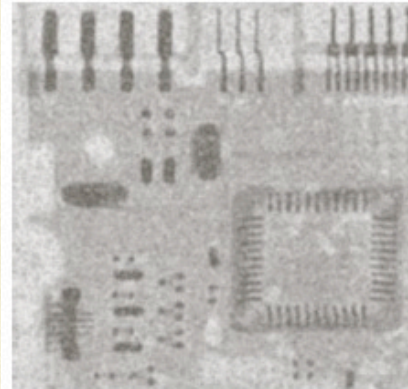
original



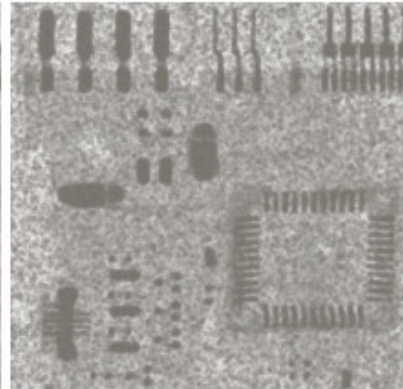
Original + salt and pepper noise



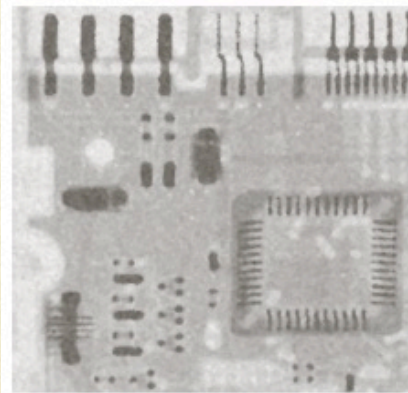
Arithmetic mean filter



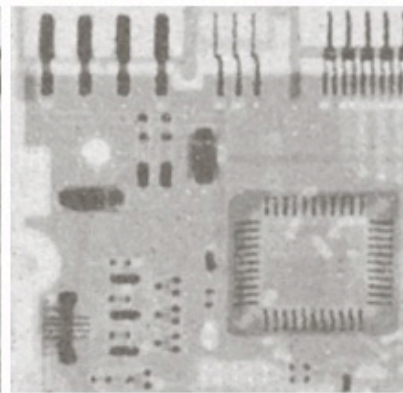
Geometric mean filter



Median filter

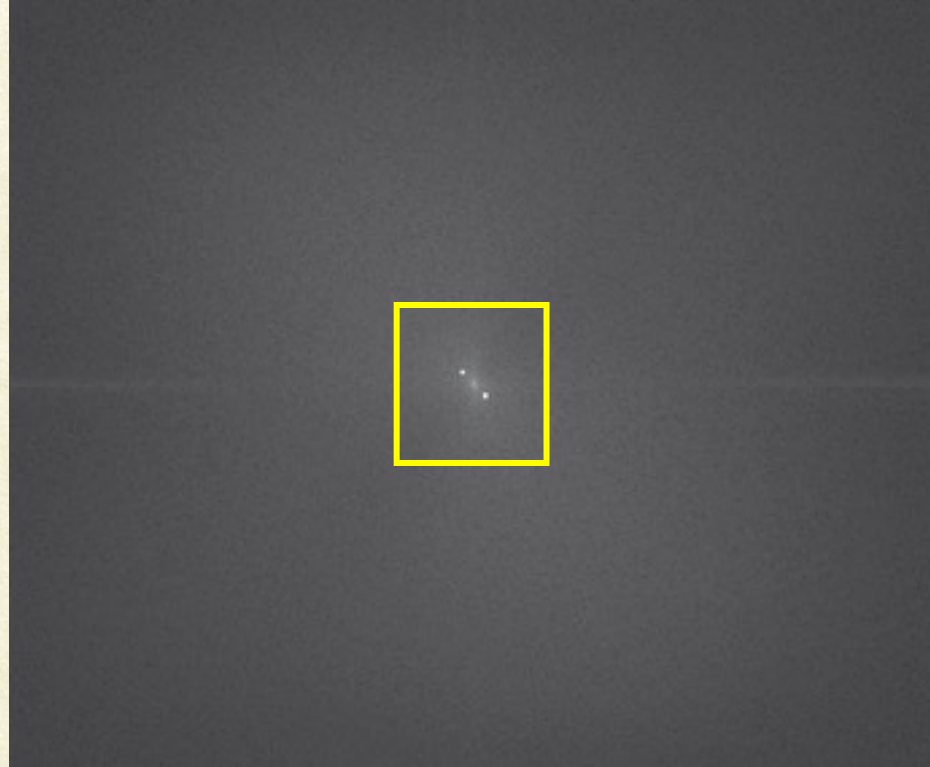


Alpha Trimmed filter





Additive Sinusoidal Noise



Additive noise of regular frequencies are easier to handle in the frequency domain.



Questions?